Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Driver 1 and matrix

File: matrix1.kicad_sch

Sunflower 4-layer density study (pn-dense-1d): 67-LED dense Penrose P3 tiling of XL-1615RGBC-RF LEDs (rhomb vertices, 2.547 mm closest pair) in a 38 mm disc: 66 on 1x AW20198QNR, the centre LED on RP2350 GPIO7/8/9; USB-C on the back, light sensor at the rim, 4
38 mm round board, Vogel-spiral LED field, 35 mm round LiPo cell behind the board.
Solid GND thermal pads on U3, U5, and all AW20198QNR QFN-40 drivers (0 via-in-pad).

Drawn by scripts/sunflower/sch_gen.py

67-LED dense Penrose P3 tiling of XL-1615RGBC-RF LEDs (rhomb vertices, 2.547 mm closest pair) i

CERN-OHL-P-2.0

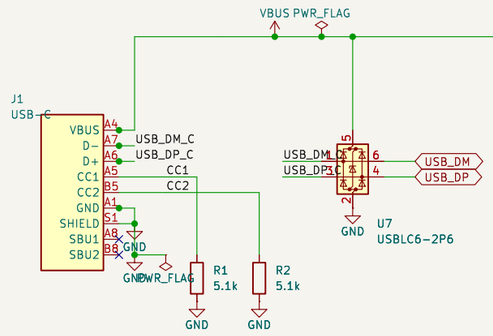
Sheet: /

File: sunflower.kicad_sch

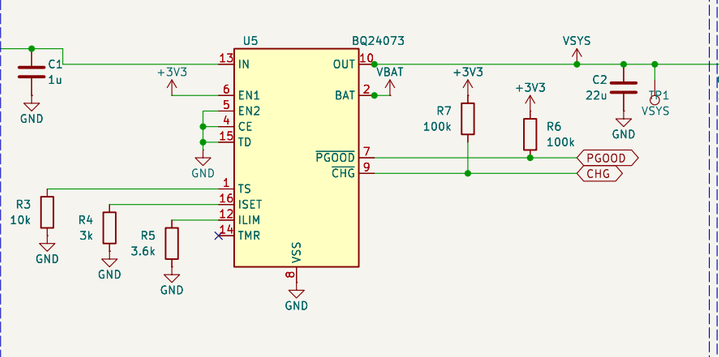
Title: sunflower (pn-dense-1d): top

Size: A4	Date: 2026-11-05	**Rev: PN1**
KiCad E.D.A. 9.0.8		Id: 1/4

USB-C, ESD

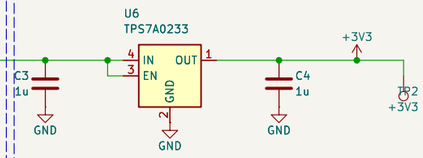


CHARGER AND POWER PATH



USB500: 450–500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

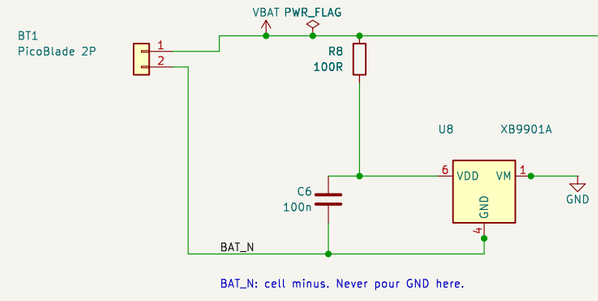
3V3



3V3 follows VSYS down: AW20198 inputs stay within VCC + 0.3 V.
150 mA max.

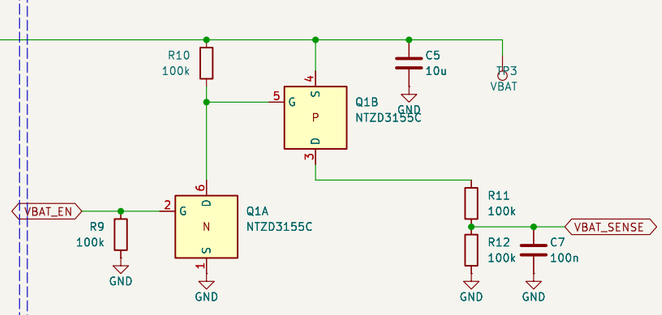
CELL AND PROTECTION

Molex PicoBlade 1.25 mm socket (the friction-fit "JST 1.25" on most small LiPos) for the 403535 (35 mm round) cell. Pin 1 = +. Check the plug's polarity: makers differ.



BAT_N: cell minus. Never pour GND here.

VBAT SENSE



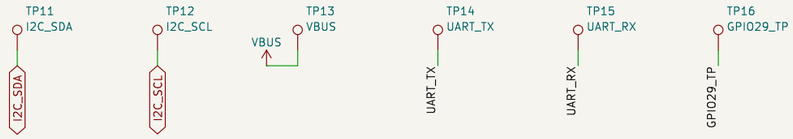
Drawn by scripts/sunflower/sch_gen.py
USB-C wires, charger, protection, LDO, VBAT sense (35 mm round cell on a PicoBlade 1.25 mm plug)
CERN-OHL-P-2.0

Sheet: /Power/
File: power.kicad_sch

Title: sunflower (pn-dense-1d): power

Size: A3	Date: 2026-11-05	**Rev: PN1**
KiCad E.D.A. 9.0.8		Id: 2/4

BRING-UP PADS: I2C0, VBUS, UART1 (GPIO24 TX, GPIO25 RX), SPARE ADC GPIO29



BOARD ID STRAPS

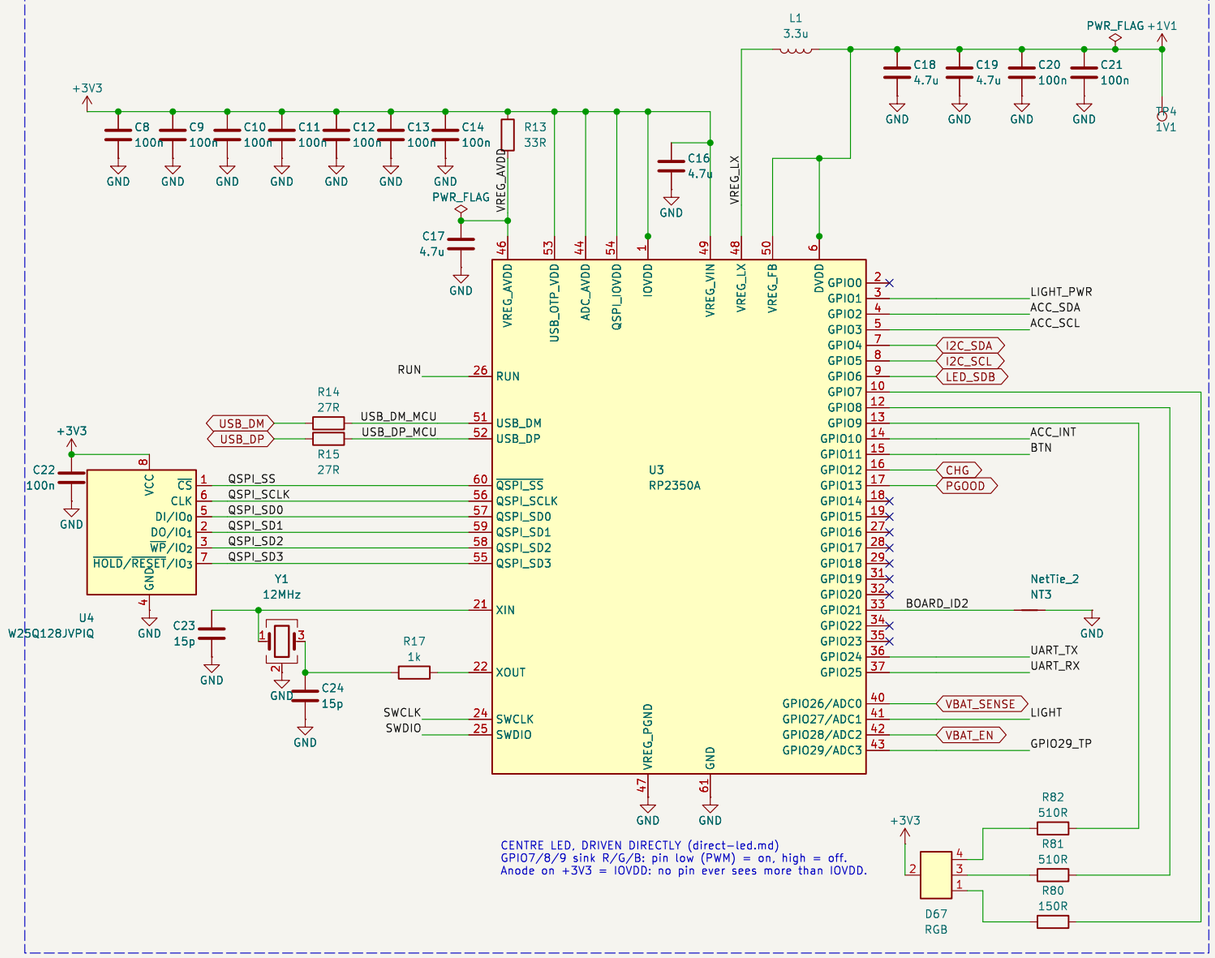
GPIO19, GPIO20, GPIO21: read with internal pull-ups.

ID = GPIO19 + 2 x GPIO20 + 4 x GPIO21; tied to GND = 0, open = 1.

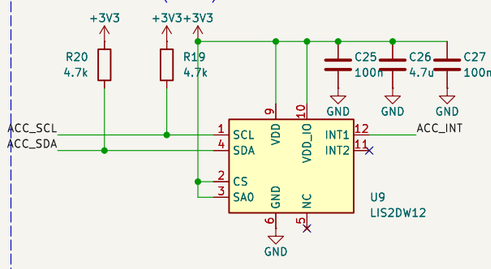
This board: GPIO21 tied (NT), ID = 3 (0b011).

Boards without straps read 7.

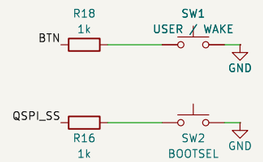
RP2350A, FLASH, CRYSTAL



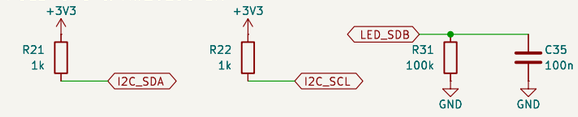
ACCELEROMETER (WAKE)



PEG-LESS SIDE BUTTONS (EVQP7J01P)



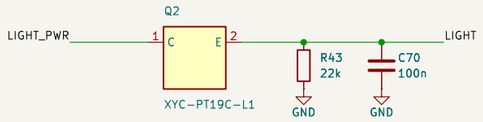
I2C0 PULL-UPS & AW20198 EN



TEST PADS



AMBIENT LIGHT SENSOR (TOP BEZEL)



Drawn by scripts/sunflower/sch_gen.py

RP2350A, flash, crystal, accelerometer, peg-less side buttons, top-bezel light sensor

CERN-OHL-P-2.0

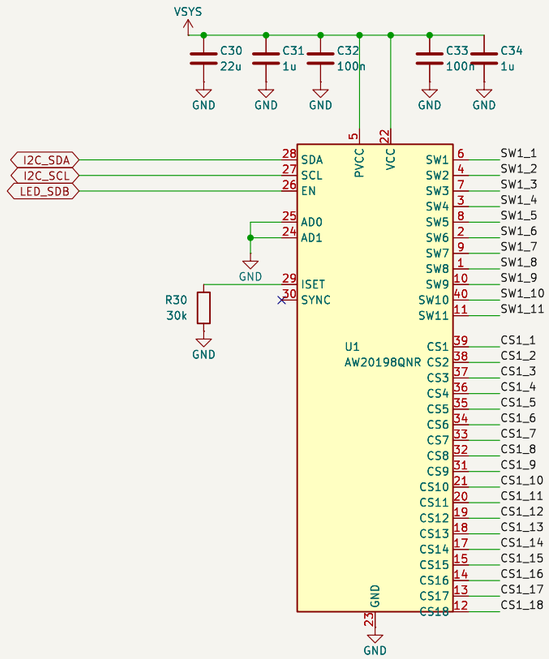
Sheet: /MCU/

File: mcu.kicad_sch

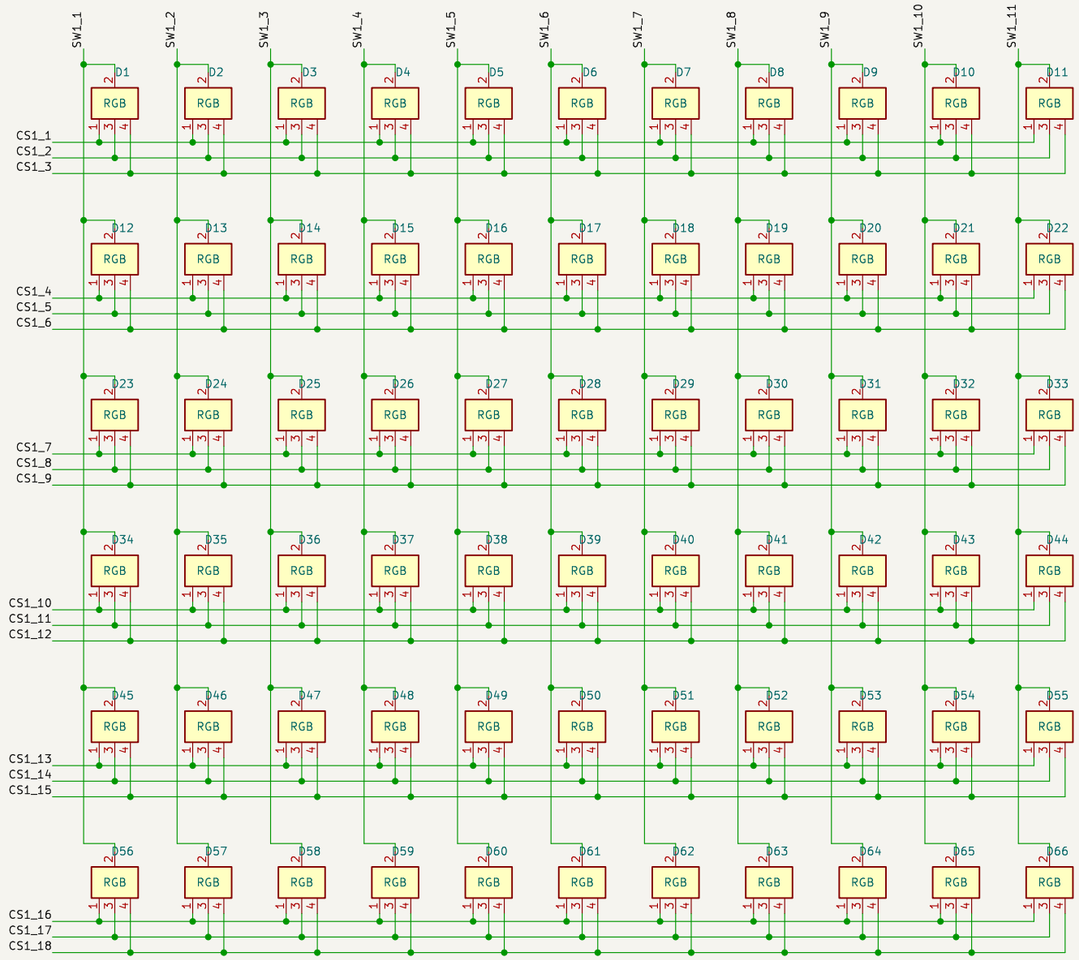
Title: sunflower (pn-dense-1d): mcu

Size: A3	Date: 2026-11-05	**Rev: PN1**
KiCad E.D.A. 9.0.8		Id: 3/4

DRIVER 1 (U1, 0x20)



66 RGB LEDS: R=CS(3k−2), G=CS(3k−1), B=CS(3k)



Drawn by scripts/sunflower/sch_gen.py
U1 AW20198QNR (0x20), 66 RGB LEDs (drivers-aw20198.md)
CERN-OHL-P-2.0

Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

Title: sunflower (pn-dense-1d): driver 1 and matrix

Size: A3	Date: 2026-11-05	**Rev: PN1**
KiCad E.D.A. 9.0.8		Id: 4/4